Ben Meyers

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PORTFOLIO — Code, Visualizations, Technical Writing — https://benmeyersusc.github.io/Portfolio

- Trainable Template Tensor Network (TTTN): C++ machine learning library, tensor shapes and architectures as *types*, neural network weights with inferred shape and automatic forward/backward + Adam states.
- Transformer Grokking: Modular addition transformer built and trained in TTTN, training and activation lenses.
- Neural Compiler: Encoder-Decoder transformer built and trained in TTTN which compiles simple C to simple assembly.
- J-Vision: Technical exposition + visualization of Anthropic's J-Lens and J-Space; proposed applications for research.
- AlphaToe: AlphaZero-inspired architecture built and trained in TTTN, used to learn Tic Tac Toe.
- Lambda Calculus Interpreter: Untyped Lambda Calculus interpreter in Python.
- TuringViz: C++ based graphical Turing Machine emulator with biological protein-based inspiration.

SKILLS AND INTERESTS

Technical: C++ | Python | Java | TensorFlow/PyTorch | LLM Harnessing | Agentic Programming | AWS/Azure | SQL
Interests: Cognition | Progressive Rock | Reading | Writing | Analog Synthesis | Evolution | Breakfast | Financial Derivatives

EDUCATION

Harvard University – Cambridge, MA *August 2026 – May 2028 (expected)*
ME in Computational Science and Engineering
Relevant Coursework: Advanced Scientific Computing: Numerical Methods, Mathematical Modeling for Computational Science, Introduction to Reinforcement Learning, Stochastic Methods in Artificial Intelligence

University of Southern California – Los Angeles, CA *August 2022 – May 2026*
Magna Cum Laude: BS in Business Administration, Minor in Computer Programming (4.0 minor GPA) GPA: 3.87 / 4.00
Relevant Coursework: Calculus, Statistics, Python, OOP in Java, Data Structures in C++, Applied Neural Networks, Advanced NLP, Business Applications of Deep Learning, Professional C++, Language Faculty Science, Human Language and Technology

EXPERIENCE

Sage Financial Group – Conshohocken, PA *May 2024 – Jan 2025 | May 2026 – Aug 2026*
AI Development Intern
- Sage Intelligence: Azure-hosted LLM platform, ~25 tools, ~5 RAG connections, rolled out to ~50 employees in August 2026
- Generalized document packages: natural language list → Python + sub-agent workflows (optional Outlook + Office add-ins)
- Designed memory systems for standard chats, Sage Projects, and user email style
- Advanced token usage and efficiency telemetry: >65% of tokens across all apps were cache reads, cutting cost rate by 50%

Bridgeforce Data Solutions – Chadds Ford, PA (Hybrid) *May 2025 – Aug 2025*
AI Development Intern
- AI Agent Assist: AWS-hosted credit dispute management system based on dozens of LLM tools performing RAG searches, document reading and writing, and natural-language-to-SQL in Redshift environment of hundreds of millions of accounts
- Rolled out to dozens of agents at a large credit furnisher by August 2025

LEADERSHIP AND INVOLVEMENT

Learning Assistant *Jan 2025 – May 2026*
Office hours, mentorship, course material development and delivery
- Data Visualization + UI Design (Spring 2026)
- Managing Data in C++ (Fall 2025, Spring 2026)
- Object Oriented Programming in Java (Spring 2025)

Neural Networks Group *Feb 2025 – May 2026*
- Co-created and led operations of reading and coding group within Undergraduate Students in Linguistics club
- Weekly lessons (given by co-creators) on neural networks and AI safety from first principles

Volunteer Research Assistant @ USC ISI *Aug 2025 – Oct 2025*
- Contributed to research project proposals, literature reviews, constructed Python spines for LLM experiments
- Topics reviewed: LLM red-teaming with adversarially fine-tuned LLMs, distillation of online communities into fine-tuned LLMs using GRPO reinforcement learning, benchmark frameworks to test LLM resistance to harmful prompts